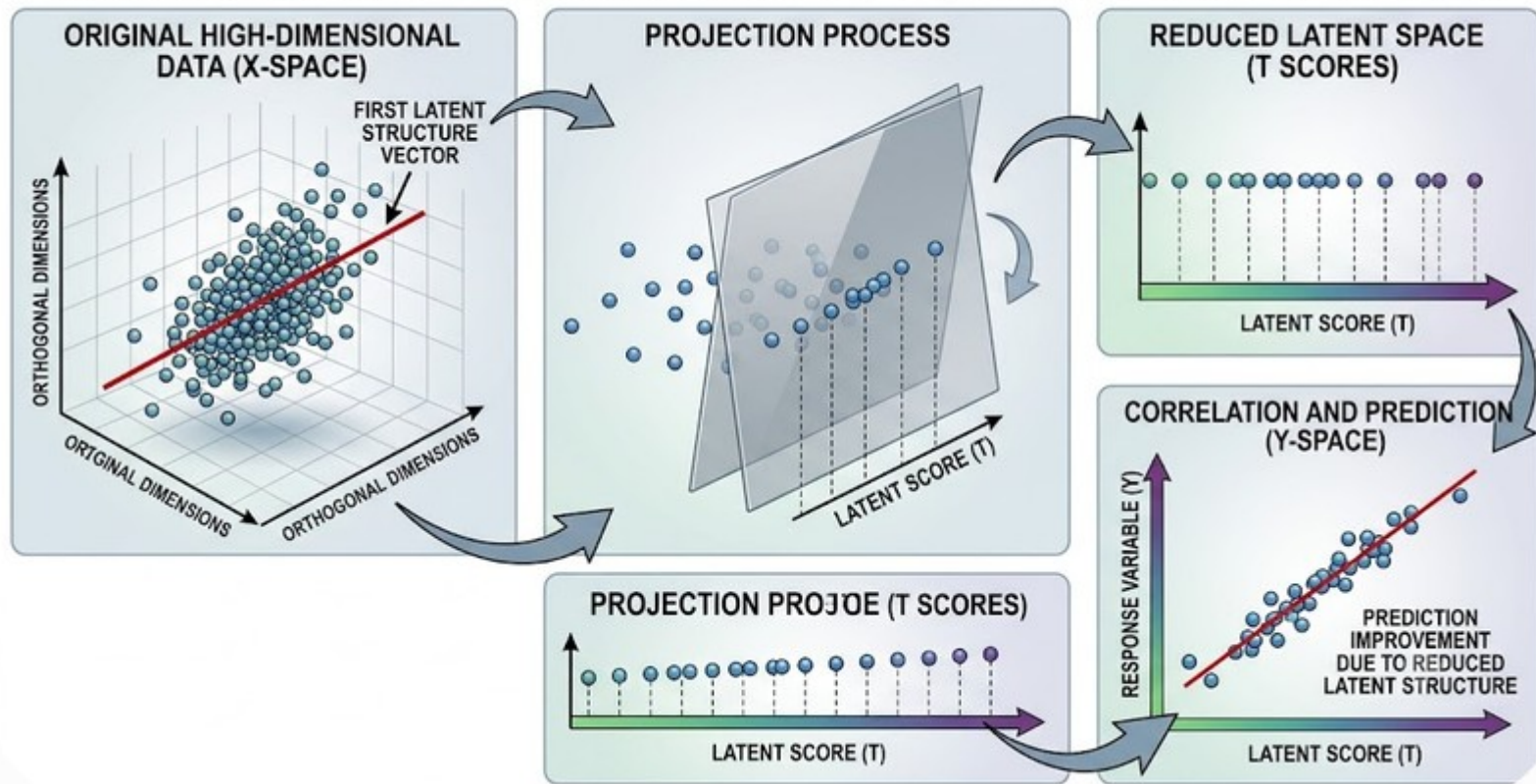


PROJECTION ONTO LATENT STRUCTURES (PLS): CONCEPT AND WORKFLOW



Mid-Term Feedback

A. Instructions (1 Unclear -10 Very Clear): 9

B. Questions (1 Unclear -10 Very Clear): 9

C. Difficulty (1 Very Difficult – 10 Very Easy): 5.5 Moderate

D. Time (1 Very less - 10 Plenty): 6

MCQ for endsem, no negative marking, home preferred, cheat sheet not allowed, discuss answers, partial marking, self brought calculator

Sir if $p \gg n$, the matrix is not square. So how can we comment on it being

"On the 'Demonstration of Variance-Covariance Principles' slide (Slide No. 2 of VarCovProblems_pyLinks.pptx), the sample problem lists a correlation coefficient value of '1.03'. If the correlation coefficient is equal to or greater than 1 (≥ 1), it should form a straight line rather than an ellipse, right? Could you kindly check this sample problem from the shared slide? Thanks in advance."

sir one of the reason it could if rows/columns is become linearly dependent when P is large kind of over-parameterize . then it can become singular

CH5440 Partial Least Squares

Ref.:

Cox, Ian and Marie Gaudard. Discovering Partial Least Squares with JMP®.

Copyright 2013, SAS Institute, Inc.

Cary, North Carolina, USA JMP Trial version software for limited period may be obtained from their website

Data Sets: <http://support.sas.com/cox> or <http://support.sas.com/audard>

Any reference inadvertently omitted will be added on intimation

Resources & Links

Textbooks:

Links: <https://community.jmp.com/t5/video/gallerypage/user-id/1670/video-id/6314798700112>

Partial Least Squares (PLS) Regression and its Robustification TU

Wien [https://repositum.tuwien.at › bitstream › Kavsek B...](https://repositum.tuwien.at/bitstream/handle/20.500.12708/10000/1/PLS%20Regression%20and%20its%20Robustification%20TU%20Wien.pdf)

<https://www.mathworks.com/help/stats/partial-least-squares-regression-and-principal-components-regression.html>

Partial Least Squares (PLS): Introduction

- ❖ Multivariate modeling technique
- ❖ Central Idea: Projection
- ❖ IV Paradigm: Data explosion is a fourth learning paradigm, the other three being
 - (i) Theory (not always tractable due to unknown parameter)
 - (ii) Experimentation (expensive, unsafe, tedious)
 - (iii) Simulation (assumptions dependent, simplified case)as means to drive new understanding (Microsoft Research 2009).

Automated Test Equipment (ATE)

- ❖ Any computer-controlled system used to automatically perform, measure, and evaluate tests on a manufactured device.
- ❖ A classic example is a Printed Circuit Board (PCB) In-Circuit Tester (ICT), which uses a "bed of nails" fixture to connect to a board, power it on, and automatically check all electrical components for manufacturing defects

Automated Test Equipment (ATE)

- ❖ ATE: Computerized hardware and software that automatically conducts tests on electronic devices to measure functionality, performance, and defects.
- ❖ Replaces manual bench instruments, ATE performs testing and diagnostics, speeding up manufacturing yields for SCs, PCBs, and electronic modules.
- ❖ **Master Controller:** Typically an industrial computer running automated test sequences
- ❖ **System Instruments:** Modules that generate test signals and measure responses (e.g., oscilloscopes, waveform generators, power supplies).
- ❖ **Mass Interconnect:** The connector interface that safely routes signals between the test instruments and the Device Under Test (DUT):

Partial Least Squares (PLS): Introduction

- ❖ Laboratories, production lines, & cars are equipped with relatively low-cost instrumentation routinely producing huge volumes of complex data
- ❖ Huge volumes of data, along with automated test equipment, results in datasets with many features (p -columns), and/or many observations (n -rows).
- ❖ Often, it is inexpensive to increase features but expensive to increase n .

Partial Least Squares (PLS): Purpose

- ❖ Clearly defined predictors and responses unlike in PCA
- ❖ PLS builds **statistical predictive** models

PLS is useful

- ❖ when $p \gg n$, features dominate in number over observations
- ❖ Tall data (when $n \gg p$, and n is large or very large)
- ❖ Square data (when $n \sim p$, and n is large or very large)
- ❖ Collinear variables: Variables that convey the same, or nearly the same information, nearly dependent columns leading to $\mathbf{X}$ matrix singularity
- ❖ Noisy data

Projection on Latent Structures (PLS)

- ❖ Recent technique that combines features from PCA and MLR*
- ❖ Its goal is to predict a set of dependent variables (Y) from a set of independent variables or predictors (X).
- ❖ This prediction is achieved by extracting from the predictors a set of orthogonal factors called latent variables which have the best predictive power

*MLR – Multiple Linear Regression

PLS: Key Aspects

- ❖ PLS regression is particularly useful when we need to predict a set of dependent variables from a (very large) set of independent variables (i.e., predictors/features).
- ❖ It originated in the social sciences (specifically economics) but became popular first in chemometrics (e.g., computational chemistry) and in sensory evaluation.
- ❖ PLS regression is also becoming a tool of choice as a multivariate technique for both non-experimental and experimental data

PLS: Goals

- ❖ The goal of PLS regression is to predict Y from X and
- ❖ To describe their common structure.
- ❖ When Y is a vector and X is a full rank matrix, this goal could be accomplished using ordinary multiple linear regression (MLR).
- ❖ When the number of predictors (p) is large compared to the number of observations (n), X is likely to be singular and
- ❖ Regression approach is no longer feasible because of multicollinearity

Common Structure

- ❖ In Partial Least Squares (PLS) regression,
"identifying the common structure between $\mathbf{X}$ and $\mathbf{Y}$ "
- ❖ means finding the underlying, hidden factors (called latent variables) that **simultaneously** explain the **variation** in the predictor variables ($\mathbf{X}$) and response variables ($\mathbf{Y}$), while **maximizing** the **connection between them**.
- ❖ Rather than looking at $\mathbf{X}$ and $\mathbf{Y}$ in isolation, PLS acts like a matchmaker, looking for the shared patterns or "common ground" that bind them together

How PLS Finds the Common Structure

- ❖ If you look at standard Multiple Linear Regression (MLR), it only focuses on predicting Y .
- ❖ Principal Component Analysis (PCA), on the other hand, only looks at X to find directions of maximum variance.
- ❖ PLS combines these ideas.
- ❖ It **decomposes** both X and Y at the same time to **capture the shared covariance between them**.

PLS decomposes X & Y Simultaneously to capture the shared covariance between them

- ❖ PLS, looks at both X and Y at the exact same time.
- ❖ It breaks both matrices down into smaller, simpler components (latent variables) together, rather than treating them as isolated islands
- ❖ **To capture the shared covariance between them:** Covariance is a measure of how much two variables change together.
- ❖ Instead of just finding what makes X vary (like PCA does), **PLS specifically hunts for the hidden patterns in X that are directly related to the hidden patterns in Y .**
- ❖ **It projects the data into a new, lower-dimensional space where the relationship (the covariance) between the X -components and Y -components is maximized.**

Hunting for Hidden Patterns

When people say "PLS specifically hunts for the hidden patterns in X and Y," they mean that **Partial Least Squares (PLS)** does not simply look at the variation in the predictor variables **X** or the response variables **Y** separately. Instead, it searches for **latent (hidden) variables** that explain **both** datasets simultaneously.

What are "hidden patterns"?

A hidden pattern is an **underlying relationship** that is not directly visible in the original variables but can be revealed by combining them.

Suppose we have

$$X = [X_1, X_2, X_3, X_4]$$

and

$$Y = [Y_1, Y_2]$$

None of the individual X_i may predict Y very well. However, a particular combination such as

$$t = 0.6X_1 + 0.5X_2 - 0.3X_3 + 0.4X_4$$

may strongly predict both responses.

This linear combination t is called a **latent score**.

The pattern is "hidden" because it does not correspond to any single measured variable.

Hunting for Hidden Patterns

What does PLS actually search for?

PLS searches for a direction in X -space

$$t = Xw$$

where

- w = weight vector
- t = latent variable (score)

such that

1. t explains a large amount of variation in X , **and**
2. t has maximum covariance with Y .

Thus PLS solves

Find w so that $\text{Cov}(Xw, Y)$ is maximum.

Why is this called "hunting"?

Imagine you have thousands of measured variables.

Most variables contain

- useful information,
- noise,
- redundant information.

PLS searches through all possible combinations until it finds the combinations that best explain Y .

It is similar to searching for the strongest signal hidden inside noisy measurements.

Example

Suppose a chemical reactor has sensors

- Temperature
- Pressure
- Flow rate
- Catalyst age

and the output is

- Product purity.

Individually,

Variable	Correlation with purity
Temperature	0.20
Pressure	0.18
Flow	0.15
Catalyst age	0.10

Each sensor alone is only weakly related to purity.

Hunting for Hidden Patterns

Hunting for Hidden Patterns

Each sensor alone is only weakly related to purity.

However,

$$0.7(\text{Temperature}) + 0.5(\text{Pressure}) - 0.4(\text{Flow}) + 0.3(\text{Catalyst age})$$

may have a correlation of **0.95** with purity.

However,

PLS discovers this hidden combination.

PCA searches for "Which direction explains the largest variation in **X**?" It ignores **Y** completely.

PCA keeps the direction where **X** varies the most, **even if that variation has nothing to do with Y**.

PLS searches for "Which direction in **X** best predicts **Y**?"

Even if that direction has only moderate variation in X, PLS prefers it if it has strong predictive power.

Hunting for Hidden Patterns

What does PLS actually search for?

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where

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PLS searches through all possible combinations until it finds the combinations that best explain Y .

It is similar to searching for the strongest signal hidden inside noisy measurements.

Hunting for Hidden Patterns

Geometric interpretation

PLS rotates the X-coordinate system until one axis points toward the direction that is **most useful for predicting Y**.

Instead of saying

"This is where X varies the most,"

PLS says

"This is where changes in X produce the largest changes in Y."

That direction is the **hidden pattern** shared by X and Y.

How PLS Finds the Common Structure

- ❖ Mathematically, it projects the high-dimensional data of both matrices into a lower-dimensional space:

$$\mathbf{X} = \mathbf{TP}' + \mathbf{E} \quad (n \times p = n \times a \times a \times p)$$

$$\mathbf{Y} = \mathbf{UQ}' + \mathbf{F}^* \quad (n \times m = n \times a \times a \times k)$$

Where:

- ❖ $\mathbf{T}$ and $\mathbf{U}$ are the score matrices (the latent variables/common structures for $\mathbf{X}$ and $\mathbf{Y}$).
- ❖ $\mathbf{P}$ and $\mathbf{Q}$ are the loadings which show how the original variables (predictors and responses) contribute to these latent structures
- ❖ $\mathbf{E}$ and $\mathbf{F}^*$ are the residuals (the noise or unexplained structure).
- ❖ The goal of PLS is to maximize the covariance between the $\mathbf{X}$ score $\mathbf{T}$ and $\mathbf{Y}$ score $\mathbf{U}$

$$\text{Max Cov}(\mathbf{T}, \mathbf{U})$$

$$\mathbf{X} = \mathbf{TP}' + \mathbf{E} \quad (\text{nxp} = \text{nxa} \times \text{axp})$$

$$\mathbf{Y} = \mathbf{UQ}' + \mathbf{F}^* \quad (\text{nxm} = \text{nxa} \times \text{axk})$$

1. A Simple Analogy Imagine you are a casting director
2. The PCA approach: You look at a group of actors (**X**) and group them solely by their heights and ages because those are the most obvious variations.
3. You completely ignore the script (**Y**)
4. PLS: You look at the actors (**X**) and the script (**Y**) simultaneously.
5. You group the actors based on traits that specifically match the emotional demands of the script.
6. In short, PLS doesn't just look for any major trends in your data;
7. It looks for the highly relevant trends that connect your inputs to your outputs.
8. This makes it incredibly powerful when you have a massive number of predictor variables (like in genomics or chemometrics) and need to predict an outcome accurately.

PLS: Fields of Application

- ❖ Psychology
- ❖ Economics
- ❖ Political science
- ❖ Environmental science
- ❖ Marketing
- ❖ Engineering
- ❖ Chemistry (organic, analytical, medical, and computational)
- ❖ Bioinformatics
- ❖ Ecology
- ❖ Biology
- ❖ Manufacturing
- ❖ Chemometrics (math, stat, and CS methods to chemistry. Design optimal experiments & extract valuable, often hidden information from complex chemical data. Highly interdisciplinary & serves as a bridge between pure chemistry and data science.

PLS: Real World Data Sets

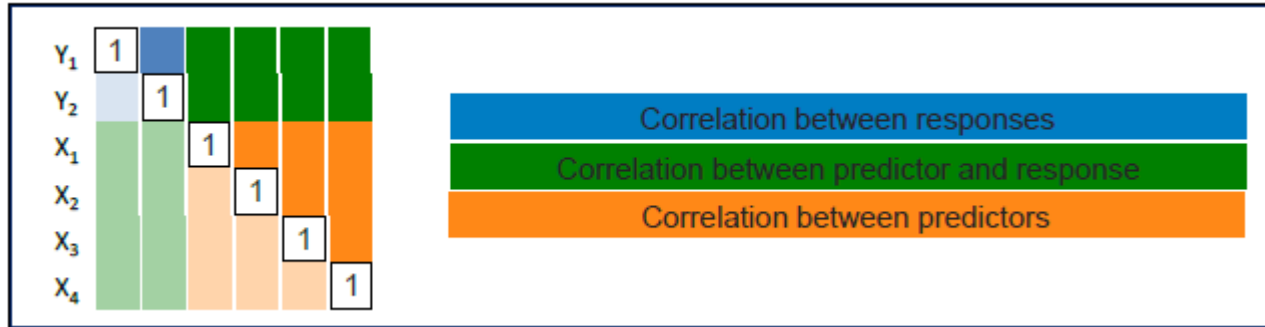
- ❖ Real world datasets frequently contain numerous, **highly correlated predictors**
- ❖ Spectroscopic data has hundreds of wavelengths, genomic studies with thousands of genes, or economic models with interconnected indicators.
- ❖ Classical regression simply cannot handle these scenarios effectively.

Need methods that embrace complexity rather than break down under it

Goal: Achieve **robust prediction** and **meaningful interpretation** despite
multicollinearity and **high dimensionality**

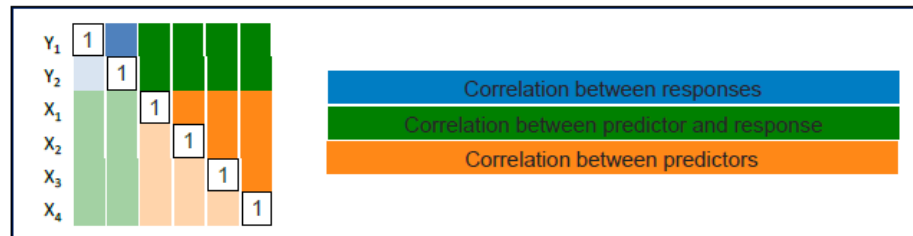
PLS: Technical Aspects

- ❖ After normalizing the data, the covariance matrix is same as the correlation matrix.
- ❖ PLS starts from the correlation matrix
- ❖ Elements of $\mathbf{E}$ (correlation matrix) should be between -1 and $+1$ and
- ❖ Matrix should be square and symmetric

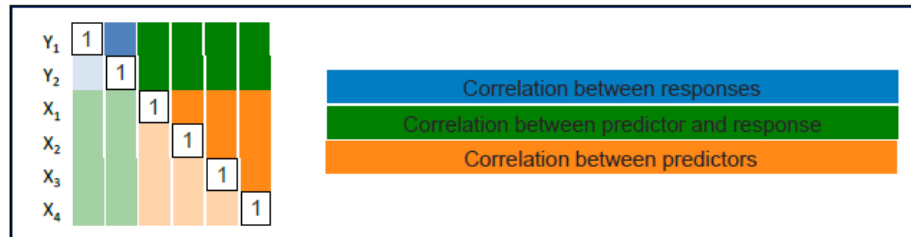


PLS: Technical Aspects

- ❖ Uses projection to reduce dimensionality by extracting factors (also called **latent variables**).
- ❖ The information it uses to do this is contained in the dark green elements of $\mathbf{E}$, here a 2×4 submatrix.
- ❖ For general values of p predictors and m measurements, this sub-matrix is not square and does not have any special symmetry.

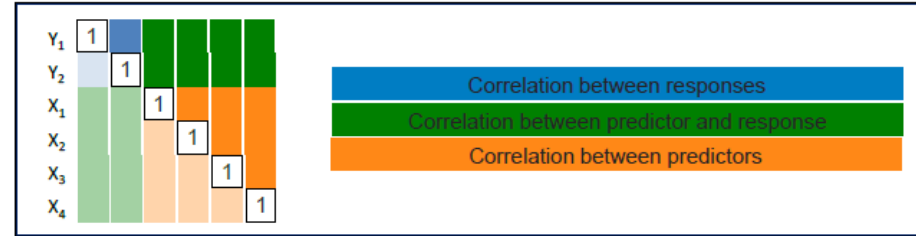


PLS: Technical Aspects



- ❖ The submatrix used by PLS contains the **correlations between the predictors and the responses**
- ❖ Using these correlations, the factors are extracted in such a way that they not only explain variation in the X and Y variables, **but they also (cor)relate the X variables to the Y variables**
- ❖ Consideration of the entire 6 x 6 correlation matrix without distinction of predictors and responses leads directly to PCA.

PLS: Technical Aspects



- ❖ Consideration of the 4 x 4 submatrix (the orange elements) leads to PCR
- ❖ The dimensionality of the $\mathbf{X}$ space is reduced through PCA, and the resulting components are treated as new predictors for each response in $\mathbf{Y}$ using MLR.
- ❖ **In many instances, though, PLS is a superior choice.**

PLS: Outcomes

- ❖ PLS not only projects the point clouds of the Xs and the Ys onto a lower-dimensional subspace,
- ❖ but it does so in a way that reflects the correlation structure between the Xs and the Ys.

PLS: Projection to Latent Structures

- ❖ Because it works by extracting latent variables, partial least squares is also called **projection to latent structures**.
- ❖ **Projection**, is fundamental to PLS
- ❖ The term “**partial least squares**” stresses the relationship of PLS to other regression methods

PLS: Projection to Latent Structures

- ❖ “Projection to latent structures” emphasizes a **more fundamental empirical principle**:

*The underlying structure of **highly dimensional data***

*associated with **complex phenomena***

is often

largely determined by a smaller number of factors or latent variables

that are not directly accessible to observation or measurement

PCA vs PLS

- ❖ PCA uses the correlation matrix for ALL variables of interest, whereas

- ❖ PLS uses the **submatrix** that links responses Y and predictors X

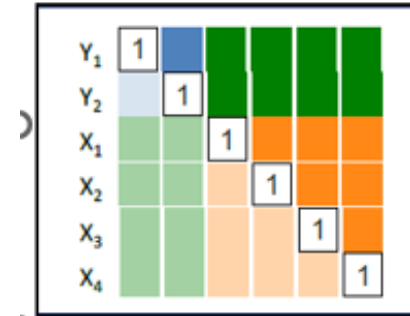
In a situation where there are both Y s and X s,

- ❖ Fig. indicates that PCA uses the orange-colored correlations, whereas

- ❖ PLS uses the green-colored correlations.

- ❖ These green entries are the correlations that link the responses and predictors.

- ❖ PLS attempts to identify factors that simultaneously reduce dimensionality and provide predictive power.



Correlation between responses

Correlation between predictor and response

Correlation between predictors

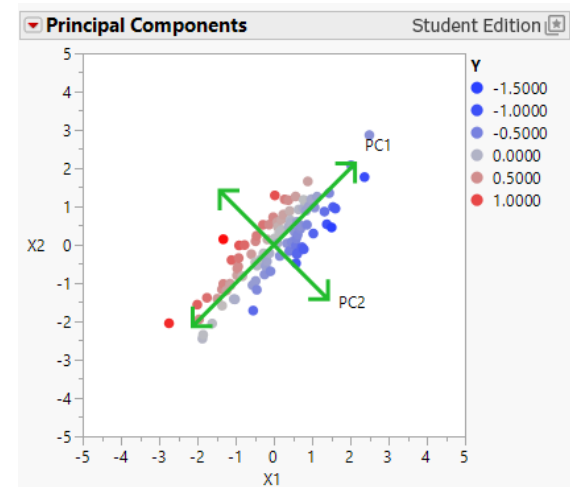
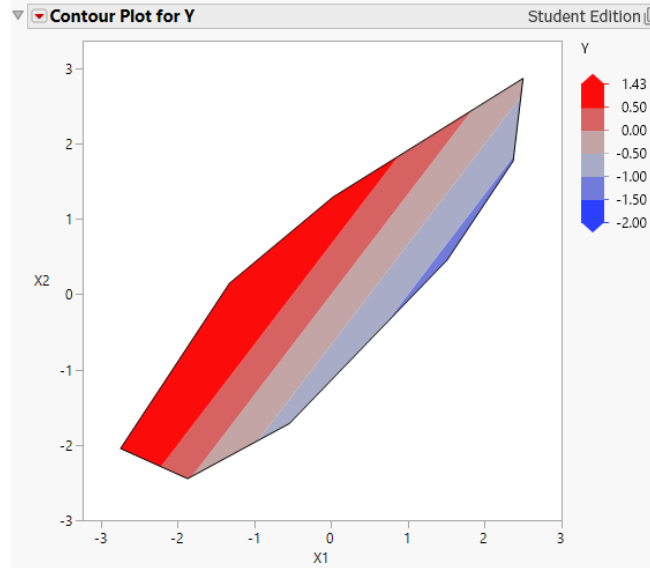
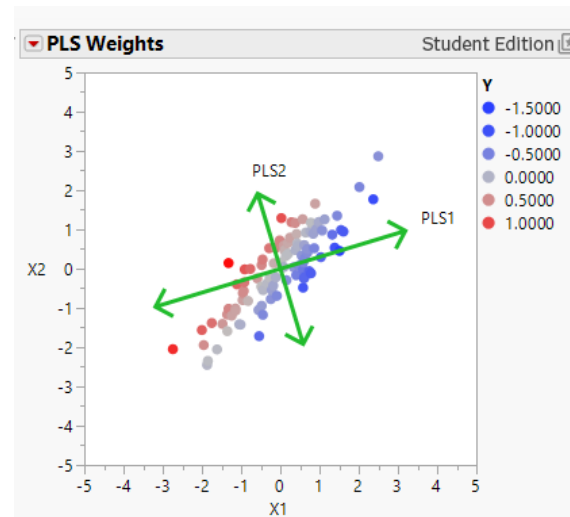
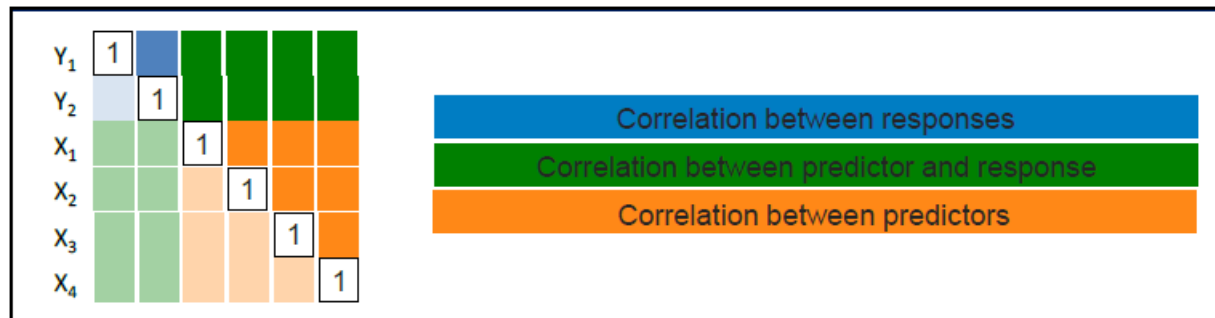
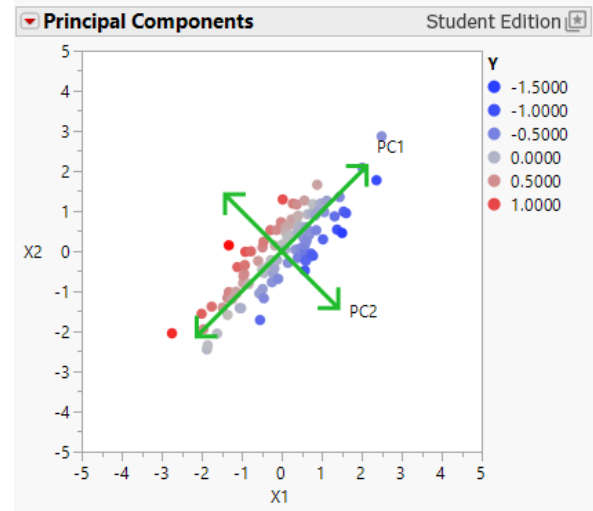
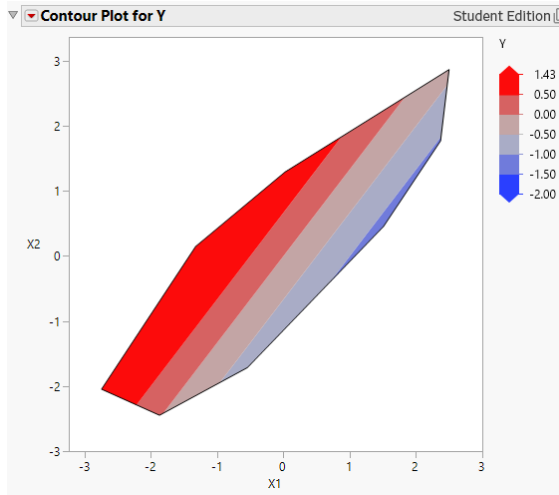


Figure 4.1: Schematic of Correlation Matrix Σ , for Y s and X s



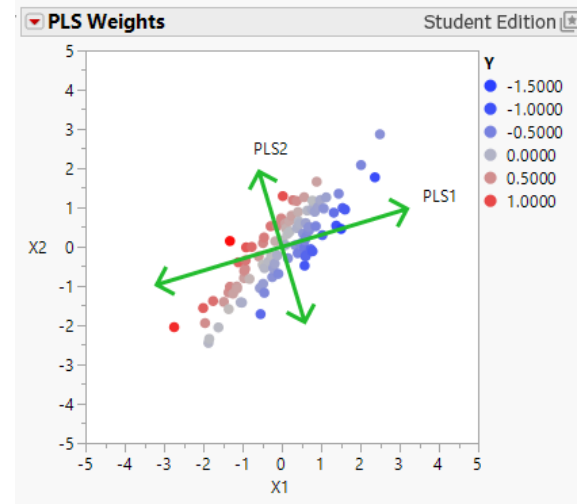
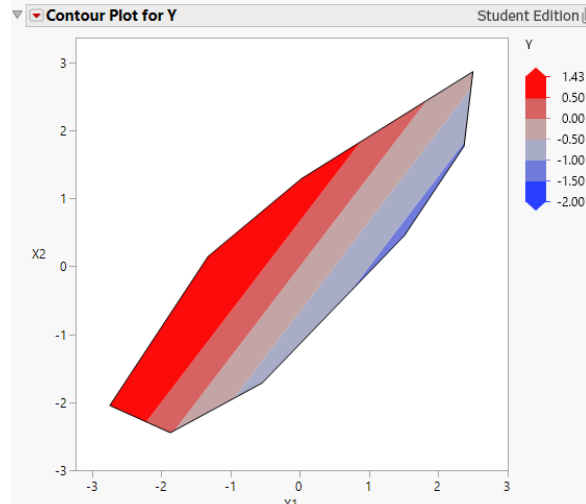
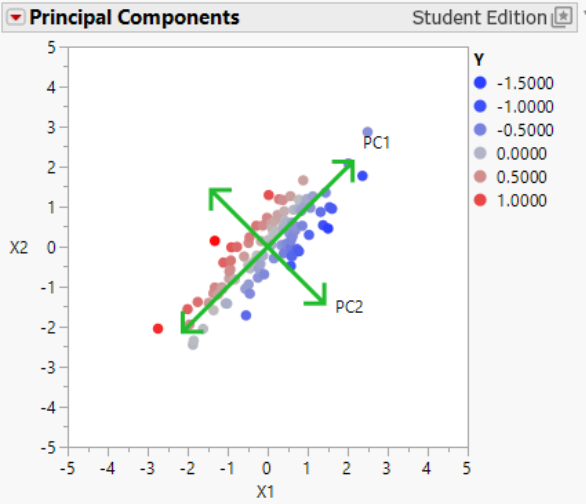
PCA vs PLS

- ❖ The PC plot shows the two principal components.
- ❖ The direction of the first component, PC1, captures as much variation as possible in the values of X_1 and X_2 regardless of the value of Y
- ❖ **PC1** is essentially perpendicular to the direction of increase in Y , shown in the contour plot.
- ❖ **PC1** ignores any variation in Y .
- ❖ The second component, PC2, captures residual variation, **again ignoring variation in Y** .



PCA vs PLS

- ❖ PLS Weights plot shows directions of 2 PLS factors, or latent variables (PLS1 and PLS2)
- ❖ PLS1 is rotated (clockwise) relative to PC1.
- ❖ PLS1 attempts to explain variation in X_1 & X_2 while
- ❖ Also explaining some of the variation in Y
- ❖ While PC1 is oriented in a direction that gives no information about Y , PLS1 is rotated slightly toward the direction of increase (or decrease) for Y



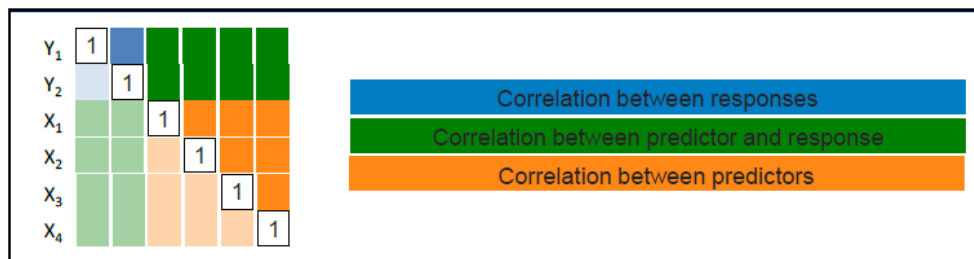
PCA vs PLS

- ❖ PCA uses the correlation matrix for ALL variables of interest, whereas
- ❖ PLS uses the **submatrix** that links responses and predictors

In a situation where there are both Ys and Xs,

- ❖ Figure indicates that **PCA uses the orange-colored correlations**, whereas
- ❖ **PLS uses the green-colored correlations.**
- ❖ These green entries are the correlations that link the responses and predictors.
- ❖ PLS attempts to identify factors that simultaneously reduce dimensionality and provide predictive power.

Figure 4.1: Schematic of Correlation Matrix Σ , for Ys and Xs



PLS: Difference With PCA

- ❖ PCA decomposes X to obtain components which best explains variation in X
- ❖ In contrast PLSR finds factors/components/latent variables from X that best predict Y

PCA vs PLS

- ❖ PCR finds components that explain variance in $\mathbf{X}$ only, without considering $\mathbf{Y}$
- ❖ Components may not be relevant for prediction
- ❖ PLS advantage is that it finds components that explain covariance between $\mathbf{X}$ and $\mathbf{Y}$ simultaneously
- ❖ Components/latent variables/factors are specifically optimized for predicting $\mathbf{Y}$
- ❖ Unlike PCR, PLS takes a fundamentally different approach to dimension reduction
- ❖ It's a supervised method where the response variable ($\mathbf{Y}$) actively guides the creation of latent components

PCA vs PLS

- ❖ PLS works by constructing a set of new latent variables (components) from the predictors ($\mathbf{X}$) that not only explain the variance in $\mathbf{X}$ but, more importantly, also maximize the covariance between $\mathbf{X}$ and $\mathbf{Y}$
- ❖ This ensures that the extracted components are maximally relevant for predicting the response, directly addressing the limitations of methods that consider $\mathbf{X}$ and $\mathbf{Y}$ separately
- ❖ This "supervised" component extraction is what gives PLS its predictive power, especially in the presence of multi-collinearity and high dimensional data.

Key Differences: PCR vs PLS

While both PCR and PLS address multicollinearity and high dimensionality, their fundamental approaches to component extraction lead to distinct advantages in different scenarios.

Component Extraction	Unsupervised; solely based on variance within predictor variables (X).	Supervised; based on maximizing covariance between X and response variable (Y).
Objective	To find components that best explain the variance in X.	To find components that best explain both the variance in X and the covariance with Y.
Y's Role	Ignored during the component extraction phase; considered only in the subsequent regression step.	Actively guides the selection and weighting of components, ensuring relevance to prediction.
Best For	Situations where understanding the internal structure of X is paramount, or when Y is not available for component selection.	When the primary goal is robust prediction of Y, especially in presence of strong multicollinearity.
Predictive Power	Potentially weaker for predicting Y, as components are not optimized for this purpose.	Typically stronger for predicting Y, due to direct optimization for the response variable.

This table highlights why PLS is often preferred in predictive modeling tasks, as its component construction inherently prioritizes the relationship with the outcome.

PLS Benefits

- ❖ **Hybrid Approach:** Combines the dimensionality reduction of PCA with the predictive power of regression
- ❖ **Focused Extraction:** Finds latent components that maximally explain the covariance between X and Y
- ❖ **Smart Reduction:** Reduces dimensionality whilst maintaining focus on predicting the response variable

PLS1: Single Response Variable

- ❖ PLS1 is designed specifically for modelling one response variable (Y) at a time.
- ❖ Algorithm extracts latent components that are optimally tuned to predict that single response.
- ❖ Simpler interpretation of results
- ❖ Component extraction focuses on a single prediction task
- ❖ Most common variant in practical applications



PLS2: Multi Response Variables

- ❖ The Multivariate Solution
- ❖ PLS2 handles multiple response variables (multiple columns in Y matrix) simultaneously.
- ❖ It extracts components that explain the covariance structure across ALL responses at once.
- ❖ Useful when responses are conceptually related
- ❖ More efficient than running separate PLS1 models
- ❖ Captures shared structure across multiple outcomes
- ❖ Most common variant in practical applications



Partial Least Squares (PLS): Pretreating Data

- ❖ First, screen and clean data for missing values, outliers and anomalies
- ❖ Works when variables involved have somewhat symmetric distributions.
- ❖ Highly skewed variables may be log-transformed prior to any analysis (e.g., log-normal distribution for skewed distributions)
- ❖ Data usually centered & scaled (normalized) before PLS analysis.
- ❖ Centering and scaling each variable results in a working data table where each variable has mean 0 and standard deviation 1.

PLS: Why Normalize?

- ❖ **Weights** that form the basis for the PLS model are **very sensitive** to the measurement units of the variables.
- ❖ Without centering and scaling, **variables with higher variance** have more influence on the model.
- ❖ **Centering and scaling puts all variables on an equal footing**
- ❖ If certain variables in **X** are indeed more important than others, and they should have higher influence, you may accomplish this by assigning them a **higher scaling weight**.

Why Pre-Process Data?

Scale Matters

- ❖ Variables measured on different scales (e.g., kilograms vs milligrams) can dominate component extraction unfairly

Centring Benefits

- ❖ Mean-centering (subtracting the mean) ensures components **represent variation** rather than absolute values

Standardization

- ❖ Scaling to unit variance
- ❖ (Auto-scaling) puts all variables on equal footing for fair comparison

Typical Preprocessing Steps

Step 1: Mean Centering

Subtract the mean from each variable in both **X** and **Y** matrices

Step 2: Scaling (Optional)

Divide by standard deviation to achieve unit variance if variables have different measurement scales

Step 3: Data Quality

Handle missing data through imputation or removal, and identify/address outliers before modelling

Interpreting Loadings and Weights

Loadings & Weights: Fundamental outputs of PLS, providing deep insights.

- ❖ Quantify contribution of each original predictor variable (X_i) to the factors
- ❖ Finding the influence of the each original predictor variable (X_i) towards prediction of the response variable(s)
- ❖ They reveal the direction and strength of the relationship between each X variable and a specific latent component which is a vector ($\mathbf{t}_i = \mathbf{X}\mathbf{w}_i$)

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Weights (W Matrix)

- ❖ Represent the importance of each predictor variable in constructing the components
- ❖ These components are specifically designed to maximize the covariance of $\mathbf{X}$ with the response variable(s) $\mathbf{Y}$
- ❖ A high absolute value for a variable in either loadings or weights indicates a strong influence on that particular latent component, making it a significant driver for the model.

Difference Between Loadings and Weights

In PLS, **weights** and **loadings** are related but they serve different purposes.

Aspect	Weight (w)	Loading (p)
Purpose	Defines the latent variable	Describes how original variables are represented by the latent variable
Computed from	Optimization of covariance between X and Y	Regression of X on the score t
Formula	$t = Xw$	$X \approx tp^T$
Interpretation	Importance of each X variable in constructing the latent score	Correlation/contribution of the latent score to reconstructing each X variable
Used for	Projection of data	Reconstruction (decomposition) of data

$$X = TP' + E \quad (n \times p = n \times a \times a \times p)$$

$$Y = UQ' + F^* \quad (n \times m = n \times a \times a \times k)$$

Difference Between Loadings and Weights

Step 1: Weight Vector

Suppose we have standardized variables X_1, X_2, X_3 .

PLS finds a weight vector

$$w = \begin{bmatrix} 0.7 \\ 0.6 \\ 0.4 \end{bmatrix}$$

and forms the first latent score

$$t = Xw$$

or

$$t = 0.7X_1 + 0.6X_2 + 0.4X_3$$

The weight vector tells us **how to combine the original variables to obtain the latent variable.**

Difference Between Loadings and Weights

Step 2: Loading Vector

Once t is obtained, PLS expresses the original X variables as

$$X \approx tp^T$$

Suppose

$$p = \begin{bmatrix} 0.9 \\ 0.8 \\ 0.5 \end{bmatrix}$$

Then

$$X_1 \approx 0.9t$$

$$X_2 \approx 0.8t$$

$$X_3 \approx 0.5t$$

The loading vector tells us **how strongly each original variable is related to the latent score.**

PLS: Theory

- ❖ A main goal of PLS is to predict one or more responses from a collection of predictors (X variables).
- ❖ This is done by extracting linear combinations of the predictors that are variously referred to as latent variables, components, or factors
- ❖ We assume that all variables are at least centered and possibly scaled
- ❖ Two general algorithms NIPALS and SIMPLS are in common use
- ❖ **NIPALS: Nonlinear Iterative Partial Least Squares**
- ❖ **SIMPLS: Statistically Inspired Partial Least Squares**